Budget-Constrained Evaluation of Open-Weight LLM Agents: Success–Budget Curves, Price Reversal, and Thinking-Budget Saturation

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Abstract

Public leaderboards rank large language model (LLM) agents by success rate under an unlimited inference budget, whereas budget-sensitive practitioners face a different question: *which model should be deployed when each task may spend at most a budget B?* We propose a budget-constrained evaluation protocol that treats the per-task budget as the independent variable and reports the success–budget curve $S(B)$ together with a family of derived metrics: the normalized area under the curve (AUBC), the budget-to-target $B_{@\tau}$, budget elasticity, and pairwise crossover points between models. We further show that for budget-unaware agents, the entire curve can be derived offline from a single maximum-budget run per task, decoupling evaluation cost from the number of budget levels. Applying the protocol to a coverage gap in existing evaluations—open-weight models served through public APIs—we evaluate 12 configurations of the DeepSeek-V4 and Qwen3 families (spanning a roughly 100-fold price range) on the three τ^3-bench customer-service domains, collecting 4,448 multi-turn tool-use trajectories. Four findings emerge. (1) Budget constraints substantially reorder the leaderboard: at 0.3× the median budget-to-50%-success, budget-constrained rankings are statistically unrelated to unconstrained ones (Kendall τ from −0.02 to 0.52), and the budget required to reach 50% success differs by 17–20× between models of comparable unconstrained accuracy. (2) The price-reversal phenomenon reported for single-turn reasoning does not appear in any of 66

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model pairs (95% upper confidence bound 4.4%), because trajectory cost is dominated by repeatedly resent context rather than thinking tokens. (3) Thinking yields a real but strongly saturating benefit: a same-backbone paired contrast is significant ($p = 0.0027$) at roughly twice the cost, yet the gain saturates at a budget of about 1K thinking tokens. (4) Open-weight hosted endpoints exhibit deterministic empty-response failures on up to 28% of tasks for one model; treating these as infrastructure errors rather than model failures would inflate its measured success by 24 percentage points. All trajectories, code, and per-token cost records are released.

Keywords: LLM agents, evaluation benchmarks, inference cost, budget-constrained evaluation, open-weight models, tool use

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1. Introduction

Large language model (LLM) agents are moving from demonstrations to deployments. In customer service, software engineering, and scientific computing, an agent completes a task through many rounds of tool calls, and its inference consumption accumulates with every step: a single agentic task can cost tens of times more than a single-turn query. For budget-sensitive deployers—precisely the primary audience of open-weight models—the operative question is not *which model is strongest* but *which model should be deployed under a per-task budget B.*

Current evaluation practice cannot answer this question. Mainstream agent leaderboards report task success as the sole headline metric, and even the most recent cost-transparent infrastructure, the Holistic Agent Leaderboard (HAL) [1], records cost *post hoc* rather than imposing it as a *prospective constraint*: it reports what each model spent without limit, not what each model could still accomplish under a cap. The distinction is not rhetorical. Per-task costs of multi-step agents are heavy-tailed—in our data, costs for the same model within one domain span an order of magnitude—so mean cost conceals how budget truncation systematically affects the expensive tail. Meanwhile, single-turn reasoning studies have documented a *price reversal* phenomenon, where models with lower list prices end up costlier due to highly variable thinking-token usage [2]; that work explicitly calls for treating inference cost as a first-class evaluation dimension and extending the analysis to agentic settings, but whether the phenomenon amplifies or vanishes over multi-step trajectories has remained an open empirical question.

This paper proposes a *budget-constrained evaluation protocol* for LLM agents. The protocol treats a hard per-task budget cap B as the independent variable of evaluation and reports the success–budget curve $S(B)$ along with a family of derived metrics: the normalized area under the budget curve (AUBC) as a scalar cost-effectiveness summary; the budget-to-target $B_{@\tau}$, i.e., the minimum budget at which success reaches τ; the budget elasticity $\varepsilon(B)$; and the set of pairwise *crossover points* between the curves of any two models, which directly yields a budget-interval-to-best-model lookup table and turns model selection into a table lookup. Rather than a one-off model comparison, the protocol is intended as a reusable, decision-theoretic evaluation instrument: any practitioner can instantiate it for their own budget, benchmark, and candidate models to obtain a deployment-ready ranking. The protocol rests on a simple but consequential observation: for an agent that is unaware of its budget, the trajectory under any budget B is exactly the prefix of its maximum-budget trajectory truncated where cumulative cost first exceeds B. Consequently, *a single maximum-budget run per task suffices* to derive $S(B)$ at every budget level offline, decoupling evaluation cost from the number of budget levels.

We apply the protocol to a coverage gap in current evaluations: *open-weight models served through public APIs*. Using the three customer-service domains of τ^3-bench [3] (airline, retail, telecom; 278 tasks in total) as the testbed, we evaluate 12 model configurations from the DeepSeek-V4 and Qwen3 families, spanning a roughly 100-fold price range, dense and mixture-of-experts architectures, and thinking and non-thinking modes, with the user simulator and judge fixed across all runs for comparability. This coverage is complementary to HAL (nine closed frontier models, no Chinese open-weight models), and for budget-sensitive deployers these models constitute the realistic candidate set.

The results support the necessity of the protocol. At 0.3× the median budget-to-50%-success, budget-constrained rankings are statistically unrelated to the unconstrained success ranking in two domains (airline: Kendall $\tau = -0.02$, $p = 0.94$; retail: $\tau = 0.17$, $p = 0.46$), recovering agreement only at 3× the median budget ($\tau = 0.60$–0.88, $p < 0.01$). To reach 50% success, the most cost-effective model (Qwen3.5-27B) needs \$0.010–\$0.015 per task while the flagship Qwen3.7-Max needs \$0.18–\$0.26—a 17–20× gap at essentially equal unconstrained accuracy. The price reversal reported for 21.8% of model pairs in single-turn settings appears in none of our 66 pairs: trajectory cost is dominated by repeatedly resent context (input-to-output token ratios of roughly 10:1 to 40:1), which dilutes the thinking-token variance that drives the single-turn effect. A thinking-budget sweep yields a clearly saturating dose–response: the benefit of thinking is confirmed by a same-backbone paired contrast ($p = 0.0027$) but saturates at about 1K thinking tokens, with a 16-fold larger budget buying only two further percentage points.

Our contributions are as follows.

1. **Protocol and metrics.** To our knowledge, the first prospective budget-constrained evaluation protocol for multi-step agents, comprising the $S(B)$/AUBC/$B_{@\tau}$/elasticity/crossover metric family, the single-run curve derivation, and a task-level paired-bootstrap inference procedure.
2. **Coverage and artifacts.** To our knowledge, the first systematic cost–capability study of Chinese open-weight agent models (12 configurations $\times$ 3 domains $\times$ 278 tasks; 4,448 valid multi-turn trajectories), with all trajectories and per-token cost records released.
3. **Findings.** Quantitative evidence that budgets reorder agent leaderboards; a mechanistic account of why single-turn price reversal disappears in multi-step tasks; a saturating dose–response curve for thinking budgets; and a systematic empty-response failure mode of open-weight hosted endpoints, with its consequences for evaluation accounting.

2. Related Work

Agent benchmarks.. τ-bench and its successor τ^3-bench [3] characterize customer-service agents through LLM-simulated users and policy-constrained tool calls; the τ^3 release fixed more than 75 task defects identified by SABER [4] and added a telecom domain, making it one of the most strictly scored multi-turn tool-use benchmarks available. SWE-bench [5], GAIA [6], and AgentBench [7] cover software engineering and general assistance, typically with ReAct-style agents [8]. Across all of these, leaderboards rank by success rate; cost appears at most as a footnote.

Cost-aware evaluation.. Kapoor et al. [9] first argued that agent evaluations neglect cost, proposed joint accuracy–cost Pareto reporting, and showed that simple baselines often Pareto-dominate elaborate agents; their follow-up HAL [1] standardized evaluation infrastructure and recorded the cost of nine closed frontier models across nine benchmarks (21,730 rollouts, $40K). Cost-of-Pass [10] defines the expected monetary cost per correct solution and tracks its frontier over model releases. The closest motivation to ours is the price-reversal study [2]: on nine single-turn reasoning datasets, 21.8% of model pairs invert between list-price order and realized-cost order, driven by thinking-token variance (up to 9.7× for a repeated query), and the authors call for cost as a first-class evaluation dimension. *Our difference from all of the above lies in the evaluation semantics*: they record or aggregate realized spending, whereas we impose a prospective budget constraint and measure the response of success to budget—the former answers "how much was spent," the latter answers "what can be accomplished under a given budget," which is the form deployment decisions take. BCAS [11] ablates design decisions of retrieval agents under fixed budgets but is restricted to one scaffold family and does not propose curve-family metrics; Efficient Agents [12] reduces agent cost by design rather than measuring cross-model budget response.

Test-time compute and thinking budgets.. Overthinking and adaptive test-time compute have been surveyed systematically [13], and adaptive effort selection has been studied for agents [14]. Public APIs now expose thinking-budget controls, turning the amount of thinking into a controllable experimental variable; existing measurements, however, focus on single-turn mathematics and coding, and the dose–response of thinking budget on multi-step agent success had not been measured systematically (our RQ3).

Open-weight and production evaluation.. Recent surveys of agent evaluation [15, 16] organize the field around behaviour, capability, reliability, and safety, and consistently note that cost efficiency and reliability remain underserved relative to task success—the gap this paper targets. Mainstream agent leaderboards moreover contain almost no Chinese open-weight models, although their roughly order-of-magnitude lower unit prices make them the de facto option for budget-sensitive deployments; production-oriented surveys likewise identify cost efficiency as a missing dimension of current evaluation frameworks [17]. This paper provides the first systematic cost–capability data for these models on a strictly scored multi-turn benchmark.

3. Budget-Constrained Evaluation Protocol

3.1. Problem Formulation

Let agent m produce trajectory $\tau(m, x)$ on task x, where the i-th model call consumes p_i input tokens and c_i output tokens (thinking tokens included, matching API billing). Given list prices $(\alpha_{\text{in}}, \alpha_{\text{out}})$ in USD per million tokens (cache-miss rates), the incremental cost of step i is $\kappa_i = (\alpha_{\text{in}}\, p_i + \alpha_{\text{out}}\, c_i)/10^6$ and the cumulative cost is $K_j = \sum_{i \le j} \kappa_i$. The outcome of task x under budget B is

$$\mathrm{succ}_B(m, x) \;=\; \mathbb{1}[\,\omega(m, x) = 1 \;\wedge\; K(m, x) \le B\,], \tag{1}$$

where $\omega(m, x) = 1$ denotes that the trajectory terminates naturally and is scored successful by the benchmark. *Accounting boundary.* The budget constrains only the evaluated agent's own calls; the user simulator and the judge are evaluation infrastructure whose costs are reported separately and excluded from B. All experiments fix the user simulator (DeepSeek-V4-Flash) and judge (DeepSeek-V4-Pro) to keep models strictly comparable.

3.2. Deriving Full Curves from Single Runs

Proposition 1 (Truncation equivalence)**.** *If an agent is budget-unaware—its policy does not condition on the remaining budget—then its trajectory under budget B equals the prefix of its maximum-budget trajectory truncated where cumulative cost first exceeds B.*

The claim follows directly from budget-independence of the policy: the per-step decision distribution is identical under both conditions, and the only difference is the external truncation time. As a corollary, $S(B)$ for every B can be computed offline from one maximum-budget run per task, making evaluation cost independent of the number of budget levels. Budget-*aware* agents violate the premise and require one run per budget level; we leave them to future work (Section 6).

3.3. Metric Family

On a per-domain logarithmically spaced budget grid (200 points spanning [0.5× the minimum completed-trajectory cost, 1.05× the maximum]) we define:

- $S(B) = \mathbb{E}_x[\mathrm{succ}_B(m, x)]$: the success–budget curve, nondecreasing in B with $S(\infty) = \text{pass@1}$;
- $\text{AUBC} = \int S\, d \log B \,/\, \Delta \log B \in [0, 1]$: a scalar summary of low-budget usability and high-budget ceiling;

- $B_{@\tau} = \min\{B : S(B) \geq \tau\}$: the budget-to-target, directly usable in deployment service-level objectives;
- elasticity $\varepsilon(B) = d \log S / d \log B$: the marginal return of budget;
- crossover points $B^*(m_1, m_2)$: budgets at which S_1 and S_2 intersect with a sign change, yielding a budget-interval-to-best-model lookup table.

Statistical inference. Within a domain all models face the same task set, so all between-model comparisons use task-level *paired* bootstrap (2,000 resamples, fixed seed) for 95% confidence intervals; rank stability across budget levels is quantified with Kendall's τ (nine tests; all significance conclusions survive Holm correction). One anchor model per provider tier is run with three independent trials to estimate run-to-run variance as a reference for interpreting single-run differences.

3.4. Research Questions

- **RQ1.** How do $S(B)$ curves differ across models, and does the unconstrained leaderboard remain informative at low budgets? Where are the crossover points?
- **RQ2.** Does the single-turn price-reversal phenomenon persist at trajectory level—do cheaper list prices still predict cheaper realized costs after thinking-token variance accumulates over many steps?
- **RQ3.** What is the dose–response of thinking budget (0/1K/4K/16K tokens) on multi-step task success and cost?

In addition to these pre-registered questions, data collection surfaced an unanticipated reliability phenomenon of open-weight hosted endpoints, which we report in Section 5.4 together with its accounting implications. Budget-aware agents, which violate the truncation-equivalence premise of Section 3.2, are left to future work.

4. Experimental Setup

4.1. Testbed

We use the three text domains of τ^3-bench [3]: airline (50 tasks), retail (114), and telecom (114), with the default `base` split and official scoring (database-state matching plus natural-language assertions). The domains represent the most common deployed agent form—multi-turn tool use under policy constraints—with largely deterministic scoring, and the τ^3 release incorporates over 75 task fixes [4]. To probe generalization beyond this task family, we conduct a validation experiment on BFCL multi-turn function calling as a second benchmark family (Section 5).

4.2. Model Matrix

We evaluate 12 configurations, all served through public Chinese APIs (Table 1 lists them): DeepSeek-V4 (flash/pro), Qwen3.7 (max/plus), Qwen3.6-flash, the open-weight Qwen3.5 series (397B-A17B, 122B-A10B, 35B-A3B, 27B), Qwen3-32B, and the same-backbone contrast pair Qwen3-235B-A22B (instruct vs. thinking). List prices span roughly two orders of magnitude (input prices from CNY 0.4 to 12 per million tokens). Qwen3-32B is run with `enable_thinking=false` due to a DashScope non-streaming restriction and billed at its non-thinking rate. The complete price table with source URLs and verification dates is released with the code (`models.yaml`). For RQ3, Qwen3.6-flash is swept over thinking budgets of 0 (disabled), 1,024, 4,096, and 16,384 tokens on the airline domain, with three trials for the 0 and 1,024 arms.

4.3. Cost Accounting and Reproducibility

Cost equals token counts times official list prices (cache-miss rates, verified 2026-07-04); CNY prices are converted at the CFETS mid rate of 6.8067 (2026-07-01). *List-price versus realized billing.* Under this accounting, the agent-side cost of all experiments is \$177.5, whereas the actual bills of the two providers total approximately CNY 825 (≈\$121); the gap stems from provider-side context-cache discounts (a 93.5% input-cache hit rate on DeepSeek, with hit prices roughly 1/120 of miss prices), limited-time promotions, and free quotas. Because such discounts vary with deployment patterns and are not reproducible, list-price accounting is our primary metric, with realized billing data released alongside. All runs use temperature 0; the user simulator and judge are fixed throughout; two judge components of the τ^3-bench harness that default to an OpenAI model were patched to DeepSeek-V4-Pro (patch released). Deterministic empty-response failures of an evaluated model are counted as task failures of that model (Section 5.4); the two empty responses of the user simulator (evaluation infrastructure) are excluded from denominators and reported. Because per-task cost is right-skewed, we report mean cost in the main tables but also release the full per-task cost distribution; for reference, medians run 5–15% below means with p90/median ratios of 1.5–1.8 (e.g., DeepSeek-V4-Flash on BFCL: mean \$0.0107, median \$0.0099, p90 \$0.0163). Per-step token and cost sequences for every trajectory, all code, and analysis scripts are released.

5. Results

5.1. *Budget Reshuffles the Leaderboard*

Table 1 reports the full results and Figure 1 the $S(B)$ curves. As a variance reference for interpreting single-run differences, the pass@1 range across three independent trials of the two anchor models is 0.08–0.10 on airline, 0.035–0.044 on retail, and 0.009–0.018 on telecom; on airline (50 tasks) between-model differences below eight points should not be over-interpreted, and the conclusions below rest on larger gaps or paired tests.

Under an unbounded budget the leading models per domain are close in pass@1 (airline 0.80–0.86, retail 0.75–0.90, telecom 0.95–1.00), yet the $S(B)$ curves show that the budget required to approach these asymptotes differs by more than an order of magnitude. By AUBC, Qwen3.5-27B ranks first or tied-first in all three domains (0.545/0.589/0.629) and DeepSeek-V4-Flash second (0.516/0.535/0.574), whereas Qwen3.7-Max—whose unbounded pass@1 matches the former—ranks last in every domain (0.160/0.144/0.143). The budget-to-target contrast is starker still: to reach 50% success, Qwen3.5-27B needs \$0.010–\$0.015 per task and Qwen3.7-Max \$0.18–\$0.26, a within-domain factor of 17 (telecom/retail) to 20 (airline).

At the rank level (full per-budget values in the released `fig3_kendall.txt`), at 0.3× the median budget-to-50% the Kendall τ between the budget-constrained ranking and the pass@1 ranking is -0.02 ($p = 0.94$) on airline and 0.17 ($p = 0.46$) on retail—*statistically unrelated*—and 0.52 ($p = 0.03$) on telecom, where the ceiling is near 1.0 and cheap models dominate across the whole budget range. Relaxing the budget to 3× restores agreement ($\tau = 0.60/0.88/0.84$, all $p < 0.01$). This "low-budget decoupling, high-budget convergence" pattern quantifies the validity boundary of the conventional leaderboard: it is informative only in the budget-abundant regime. Because the airline domain (50 tasks) has limited precision, we treat single-domain orderings as *indicative* and rest the claims on cross-domain agreement and paired tests. In the budget-interval-to-best-model lookup table, Qwen3.5-27B and

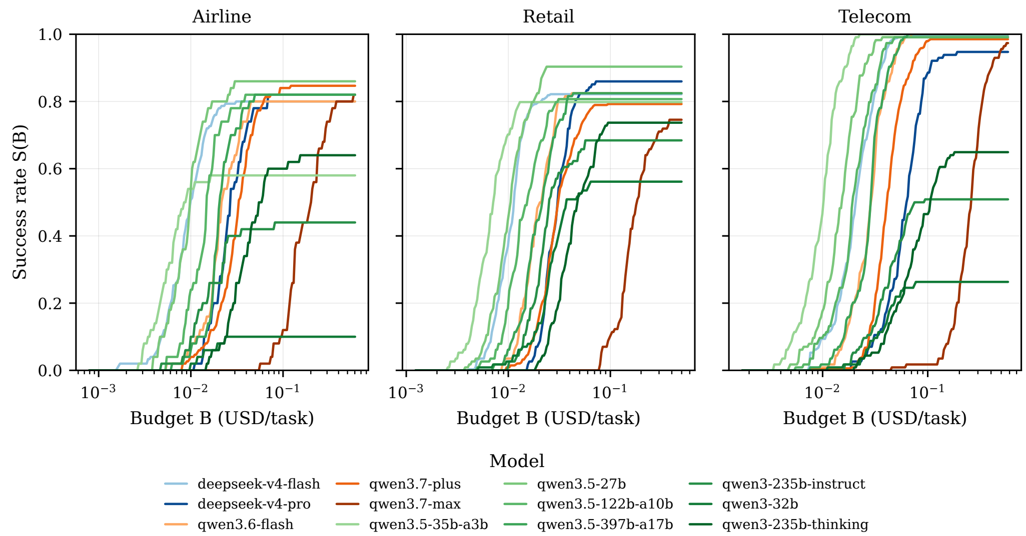

Figure 1: Success–budget curves $S(B)$ for the 12-model matrix across the three τ^3-bench domains. The x-axis is per-task budget (USD, log scale); colours group model families. Low-priced open-weight models (green) rise earliest.

DeepSeek-V4-Flash are the two budget-efficiency leaders across the three domains; the exact top spot is sensitive to cost accounting (Appendix Appendix B), whereas the robust conclusion is that no flagship model enters the top two at any budget in any domain. The AUBC-versus-price scatter (Figure 2) shows a Pareto frontier composed exclusively of low-priced open-weight models.

5.2. Absence of Price Reversal in Multi-Step Tasks

On single-turn reasoning, prior work reports that 21.8% of model pairs invert between list-price order and realized-cost order, driven by thinking-token variance. Testing all 66 model pairs by "composite list-price order versus measured per-task-cost order" (Figure 2), we find *zero inversions* (95% upper confidence bound 4.4%, Clopper–Pearson): on this testbed the list-price order predicts the realized-cost order exactly.

The mechanism lies in the cost structure. Each step of a multi-turn tool call resends a growing context, so cumulative input tokens per task reach 33K–91K while output (thinking included) tokens are only 2K–8.4K, an input-to-output ratio of roughly 10:1 to 40:1. The thinking-token variance that dominates single-turn cost is diluted at the trajectory level by (i)

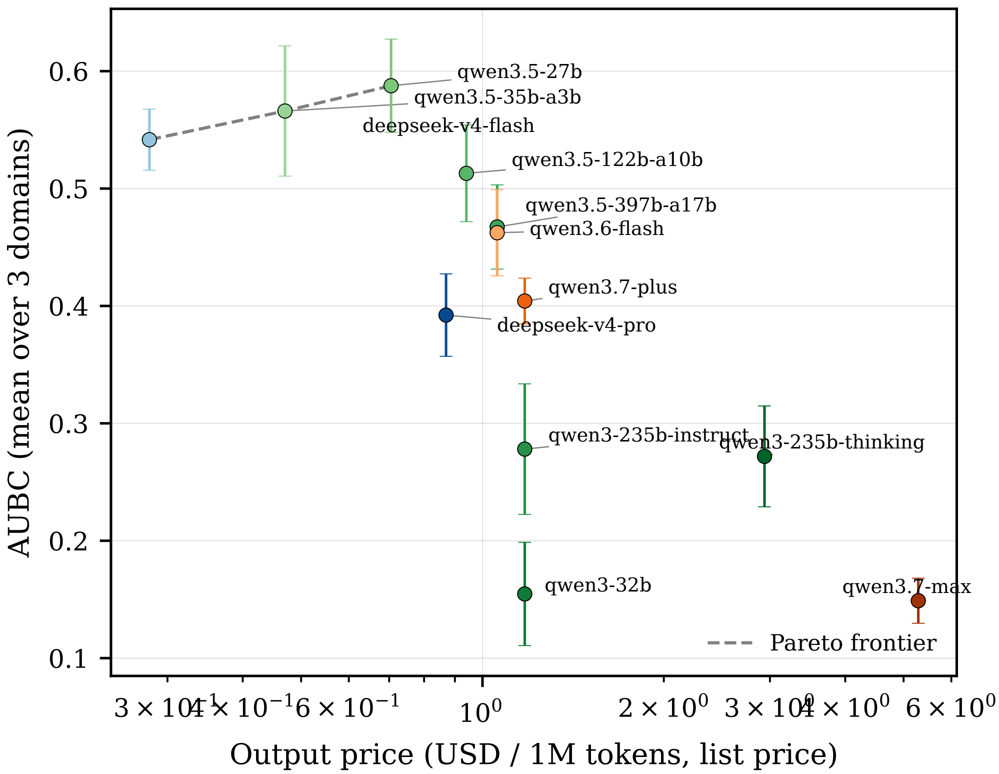

Figure 2: AUBC (mean over three domains) versus output list price (USD per million tokens, log scale). The Pareto frontier is formed exclusively by low-priced open-weight models; the flagship Qwen3.7-Max sits at the lower right (high price, low AUBC).

the large base of input cost and (ii) averaging over roughly 8–35 calls. This directly answers the call in prior work to extend cost evaluation to agentic settings: *the central cost risk of single-turn studies (thinking variance) does not appear to carry over to multi-step deployment on our benchmarks; instead, the dominant cost driver shifts to context management.*

5.3. Thinking-Budget Dose–Response

The value of thinking is characterized by two complementary pieces of evidence. *Efficacy*: on the same-backbone contrast, the thinking variant of Qwen3-235B improves over the instruct variant by +20 points on airline (0.64 vs. 0.44), +5 on retail, and +14 on telecom; a paired McNemar test over the 278 tasks gives a flip ratio of 70:38 ($p = 0.0027$)—the benefit of thinking is statistically significant on a fixed backbone, at roughly twice the cost. *Saturation*: sweeping the thinking budget of Qwen3.6-flash on airline (three trials per arm, 147 paired tasks), pass@1 is 0.762 with thinking disabled, rises to 0.823 at a 1K budget (+6.1 points at +7% cost; paired flips 20:11, $p = 0.15$, direction consistent across all three trials but not individually significant), is unchanged at 4K, and reaches only +2 further points at 16K for +24% cost. The combined deployment rule is clear: *thinking is worth buying, but only the first ∼1K tokens; larger budgets yield no measurable success gain on this benchmark.*

An unanticipated secondary finding: enabling thinking *lowers* cumulative input tokens per task from 91K to 78K—thinking lets the dialogue converge in fewer turns, so its cost is partly self-offsetting through shorter trajectories. This further explains the limited effect of thinking variance on trajectory cost observed in Section 5.2.

5.4. Reliability Failures of Open-Weight Endpoints

On 28% of airline tasks (14/50), across four in-process retries and six independent re-run rounds, Qwen3.5-35B-A3B *deterministically* returns empty responses (an assistant message with neither content nor tool calls; one further case occurs on retail); the behaviour is bound to task content rather than random. If, following common practice, these are labelled infrastructure errors and dropped from the denominator, the model's airline pass@1 is 0.818; counted as failures of the model itself (our convention), it is 0.580—*the choice of accounting convention alone shifts the measured success rate by 24 percentage points*. As a control, empty responses on the user-simulator side (infrastructure) number only two (Qwen3.7-Plus/telecom) and are excluded from denominators and reported. Separately, Qwen3-32B must be run with thinking disabled to function under DashScope's non-streaming interface. Figure 3 confirms that list-price order tracks realized-cost order (Section 5.2). These phenomena indicate that open-weight hosted endpoints exhibit systematic behavioural defects beyond rate limiting; consistent with emerging reliability-science perspectives that look beyond pass@1 [18], a budget-constrained evaluation must treat "does the model reliably return a parseable response" as a first-class part of its accounting, lest the cost-effectiveness of the cheapest models be systematically overstated.

Table 1: Main results across the 12-model matrix and three τ^3-bench domains. pass@1 is the unbounded success rate; mean cost is agent-side USD per task at list price; AUBC is the normalized area under $S(B)$ with a task-level bootstrap 95% CI; $B_{@50}/B_{@80}$ are the budgets (USD) to reach 50%/80% success ("–" if unattained). AVERAGE rows average over domains.

Model	pass@1	Mean cost (USD)	AUBC [95% CI]	$B_{@50}$ (USD)
qwen3.5-27b	0.918	0.0127	0.588 [0.546, 0.625]	0.0120
qwen3.5-35b-a3b	0.793	0.0081	0.566 [0.510, 0.621]	0.0086
deepseek-v4-flash	0.872	0.0143	0.542 [0.515, 0.568]	0.0141
qwen3.5-122b-a10b	0.873	0.0174	0.513 [0.470, 0.553]	0.0167
qwen3.5-397b-a17b	0.881	0.0248	0.467 [0.429, 0.501]	0.0239
qwen3.6-flash	0.875	0.0246	0.462 [0.425, 0.498]	0.0238
qwen3.7-plus	0.875	0.0366	0.404 [0.384, 0.423]	0.0353
deepseek-v4-pro	0.876	0.0416	0.392 [0.356, 0.426]	0.0404
qwen3-235b-instruct	0.544	0.0339	0.278 [0.223, 0.334]	0.0500
qwen3-235b-thinking	0.675	0.0615	0.272 [0.228, 0.314]	0.0695
qwen3-32b	0.308	0.0327	0.155 [0.112, 0.200]	–
qwen3.7-max	0.846	0.2124	0.149 [0.130, 0.168]	0.2162

5.5. Cross-Benchmark Generalization

To test whether the above conclusions extend beyond the customer-service task family, we replicate the protocol on a second, structurally different benchmark: the Berkeley Function-Calling Leaderboard (BFCL) [19] multi-turn function calling (`multi_turn_base`), evaluating the same 12 models on a fixed prefix of 100 tasks. Because BFCL exposes per-task token totals, each task is treated as a single-point trajectory and the same $S(B)$/AUBC pipeline applies.

The budget-efficiency ordering transfers strongly. Across the 12 shared models, the Kendall rank correlation between the τ^3-bench mean AUBC and the BFCL AUBC is $\tau = 0.758$ ($p = 2 \times 10^{-4}$; model-level bootstrap 95% CI $[0.48, 0.93]$; Figure 4). Resampling the 100 BFCL tasks (task-level bootstrap) keeps the correlation stable at $\tau = 0.774$, 95% CI $[0.67, 0.88]$, and a 50-task subsample gives the same range (Appendix Appendix B), so the result is not an artifact of the fixed first-100-task selection; the pass@1 rank correlation is weaker ($\tau = 0.504$, $p = 0.023$), consistent with our thesis that budget-conditioned metrics are more stable and more decision-relevant than raw success. The same models lead on both benchmarks: DeepSeek-V4-

Flash attains 0.93 accuracy at \$0.011 per task on BFCL (rank 1 by AUBC) and Qwen3.5-27B ranks second, while the flagship Qwen3.7-Max again ranks near the bottom (0.64 accuracy at \$0.127 per task). That a leaderboard reshuffle and a cheap-model-dominated Pareto frontier reproduce across two benchmarks with entirely different interaction structures indicates the effect is a property of budget-conditioned evaluation rather than of any single task family.

6. Discussion and Limitations

Implications for practitioners.. On our testbeds, (i) a "27B-class open-weight model with ample budget headroom" outperforms a "flagship model on a tight budget" across almost every budget interval; (ii) the thinking budget should be set to $\sim$1K rather than left unbounded, as additional thinking yields no measurable gain on these benchmarks; and (iii) model selection must be re-evaluated under the same accounting that deployment uses, since the unbounded leaderboard has no predictive power ($\tau \approx 0$) in the low-budget regime.

Threats and limitations.. (i) *Benchmark families*: although the cross-benchmark result (Section 5.5) covers two structurally different families, both are English-language and API-served; extension to software-engineering or web-navigation agents remains to be verified. (ii) *Judge dependence*: the user simulator and NL-assertion judge are fixed to DeepSeek models, a theoretical self-preference risk—but the primary judge signal is a deterministic database-state comparison, the NL assertion is auxiliary, and all evaluated models share the same judge, so any systematic bias does not affect between-model ordering. (iii) *Accounting approximation*: list prices are taken at the first tier of tiered pricing (per-task input never exceeds the tier), and cache discounts are excluded from the primary metric (Section 4.3 gives the realized-billing comparison). (iv) *Endpoint opacity*: quantization and speculative decoding of hosted endpoints are uncontrollable, so the empty-response phenomenon (Section 5.4) may stem from endpoint configuration rather than the weights themselves; the conclusions should be read as "an evaluation of that hosted endpoint" rather than "of the weights." (v) *Budget-aware agents* are out of scope: whether an explicit budget signal improves $S(B)$ is a natural follow-up requiring a dedicated agent and per-budget-level runs. (vi) *Sample size and power*: the minimum detectable effect (two proportions, $\alpha = 0.05$, power

0.8) is 28.0 pp on airline (50 tasks) and 18.6 pp on retail/telecom (114 tasks); the headline Qwen3.5-27B-vs-Qwen3.7-Max gaps (35–50 pp) are powered in all three domains, whereas smaller single-domain orderings are treated as indicative (Appendix Appendix B).

7. Conclusion

We proposed a budget-constrained evaluation protocol for LLM agents that turns the deployer's real question—which model to run under a per-task budget—into a measurable, table-lookup object: the $S(B)$ curve family and its derived metrics, together with a single-run derivation that makes the whole family cheap to obtain. An empirical study of 12 open-weight and Chinese-API models over 4,448 multi-turn trajectories yields four results: budget constraints statistically decouple the leaderboard from the conventional ranking (low-budget $\tau \approx 0$, budget-to-target differing 17–20×); the single-turn price-reversal phenomenon vanishes in multi-step tasks because the cost structure inverts; the benefit of thinking saturates at about 1K tokens; and open-weight hosted endpoints exhibit deterministic failures that must enter the evaluation accounting. A second benchmark confirms that the budget-efficiency ranking generalizes (Kendall $\tau = 0.758$, $p < 0.001$). The protocol, code, all trajectories, and per-token cost records are released, and the framework extends directly to new models, new benchmarks, and budget-aware agents.

Declarations

Ethics. This study involves no human participants, identifiable personal data, or animal experimentation; it uses public benchmarks, simulated users, and public API models, so institutional review, informed consent, and animal-welfare approval are not applicable.

Funding. The author received no specific funding for this work.

Competing interests. The author declares no competing interests.

Data availability. All trajectories, per-token cost records, the price table with sources, and analysis code are released at `https://github.com/shenzxc/budget-constrained-agent-eval`; the large raw τ^3-bench simulation logs are archived separately (DOI to be assigned on acceptance).

Generative AI use. AI-based coding and language tools were used to assist implementation and manuscript preparation; all experimental design, analysis, and scientific claims are the author's own and were verified against the released data.

Appendix A. Full per-domain results

Appendix B. Robustness analyses

We summarize three robustness checks; scripts and JSON outputs are released.

Subset sensitivity of the cross-benchmark correlation.. Resampling the 100 BFCL tasks with replacement (2,000 reps) yields a cross-benchmark AUBC Kendall τ of 0.774 (95% CI $[0.67, 0.88]$); a 50-task subsample without replacement gives the same range. The lower bound stays ≥ 0.67 in every scheme, so the positive cross-benchmark correlation does not depend on the fixed first-100-task prefix.

Cost-accounting robustness.. Recovering realized/list-price ratios from actual provider bills gives 0.135 and 0.225 for the two DeepSeek models (a structural $\sim$93% context-cache hit rate), 0.50 and 0.80 for Qwen3.7-Max/Plus (site-wide time-limited promotions), and 1.0 for the remaining eight Qwen models. Re-ranking by realized cost preserves the ordering strongly (Kendall $\tau = 0.727$, $p = 5 \times 10^{-4}$, for both AUBC and $B_{@50}$). The conclusion that no flagship model reaches the top two is unaffected; the only material change is that DeepSeek-V4-Flash overtakes Qwen3.5-27B for the single top spot, because DeepSeek's cache discount is deeper than Qwen's promotional discount. We therefore report the two as co-leaders in the main text rather than asserting a single winner.

Statistical power.. The minimum detectable effect (two independent proportions, $\alpha = 0.05$, power 0.8, conservative $p = 0.5$) is 28.0 pp for airline ($n = 50$) and 18.6 pp for retail and telecom ($n = 114$). The Qwen3.5-27B-vs-Qwen3.7-Max AUBC and pass@1 gaps (35–50 pp) exceed the MDE in all three domains and are thus powered; the thinking-budget $0 \rightarrow 1024$ contrast (6.1 pp) falls below its MDE (10.6 pp), consistent with the direction-consistent-but-not-significant reading in Section 5.3. Our McNemar implementation reproduces the main-text p-values (0.0029 vs. 0.0027; 0.151 vs. 0.15).

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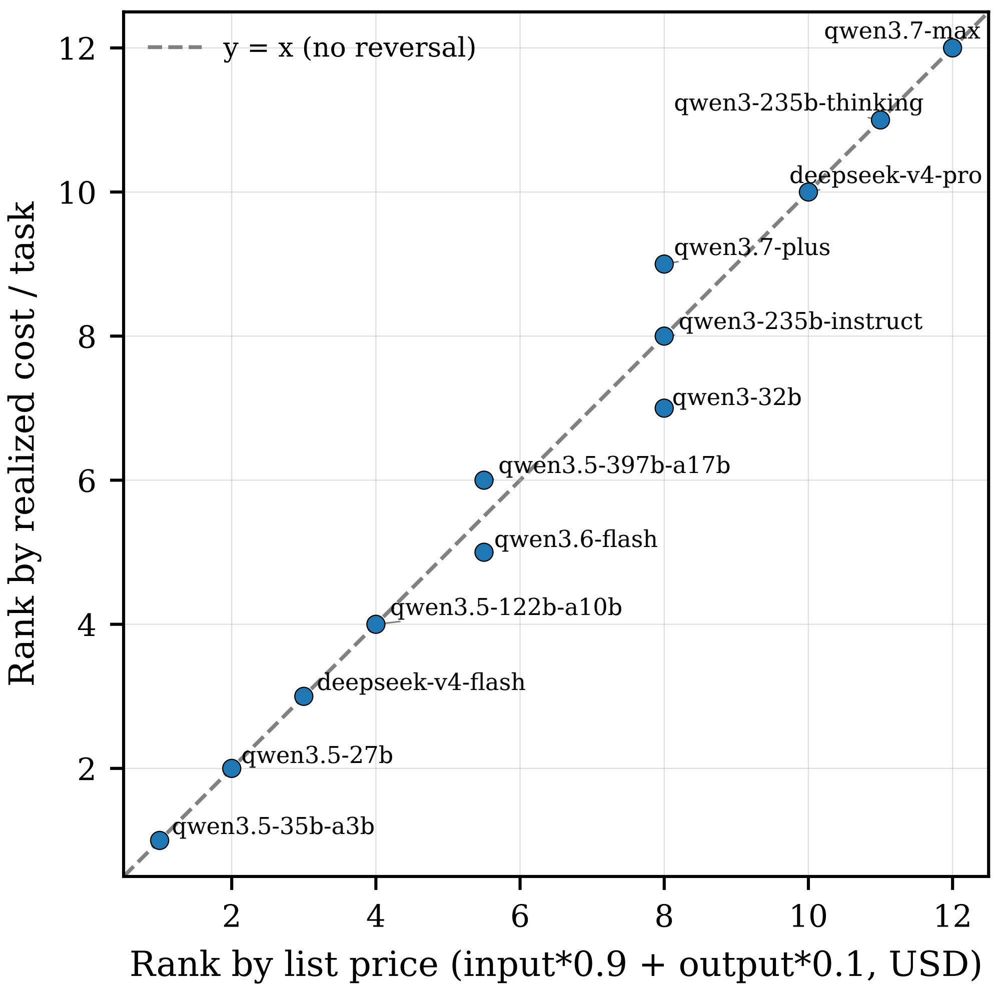

Figure 3: List-price rank versus measured per-task-cost rank across models. Points lie on the diagonal (zero of 66 pairs inverted), i.e. no price reversal in the multi-step setting.

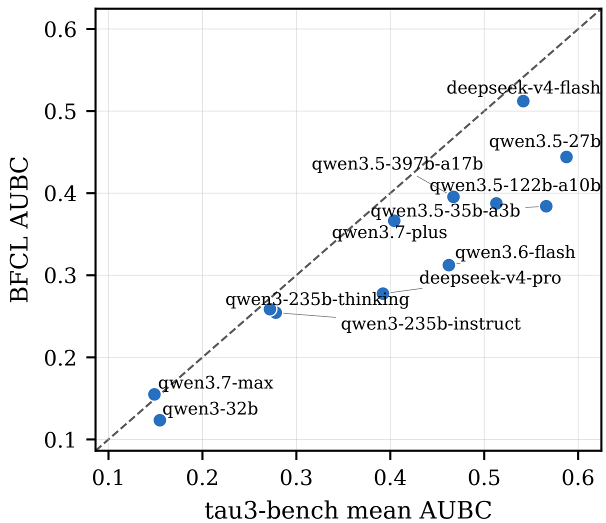

Figure 4: Cross-benchmark generalization: per-model AUBC on τ^3-bench (mean over three domains) versus BFCL `multi_turn_base`. Budget-efficiency rankings agree strongly (Kendall $\tau = 0.758$, $p < 0.001$, $N = 12$); the dashed line is the identity.

Table A.2: Full results across the 12-model matrix and three τ^3-bench domains (per-domain and per-model average). Columns as in Table 1, with $B_{@80}$ additionally reported.

Model	Domain	n_{valid}	pass@1	Mean cost (USD)	AUBC [95% CI]	$B_{@50}$ (USD)	$B_{@80}$ (USD)
deepseek-v4-flash	airline	150	0.800	\$0.0101	0.516 [0.473, 0.557]	\$0.0100	\$0.0310
deepseek-v4-flash	retail	342	0.822	\$0.0105	0.535 [0.509, 0.562]	\$0.0115	\$0.0210
deepseek-v4-flash	telecom	342	0.994	\$0.0223	0.574 [0.564, 0.583]	\$0.0207	\$0.0311
deepseek-v4-flash	AVERAGE	278	0.872	\$0.0143	0.542 [0.515, 0.568]	\$0.0141	\$0.0277
deepseek-v4-pro	airline	50	0.820	\$0.0303	0.393 [0.338, 0.444]	\$0.0272	\$0.0691
deepseek-v4-pro	retail	114	0.860	\$0.0301	0.408 [0.376, 0.438]	\$0.0311	\$0.0461
deepseek-v4-pro	telecom	114	0.947	\$0.0642	0.375 [0.352, 0.396]	\$0.0628	\$0.0892
deepseek-v4-pro	AVERAGE	92.7	0.876	\$0.0416	0.392 [0.356, 0.426]	\$0.0404	\$0.0681
qwen3-235b-instruct	airline	50	0.440	\$0.0248	0.241 [0.166, 0.320]	–	–
qwen3-235b-instruct	retail	114	0.684	\$0.0260	0.357 [0.311, 0.403]	\$0.0252	–
qwen3-235b-instruct	telecom	114	0.509	\$0.0508	0.236 [0.191, 0.280]	\$0.0748	–
qwen3-235b-instruct	AVERAGE	92.7	0.544	\$0.0339	0.278 [0.223, 0.334]	\$0.0500	–
qwen3-235b-thinking	airline	50	0.640	\$0.0522	0.267 [0.207, 0.322]	\$0.0547	–
qwen3-235b-thinking	retail	114	0.737	\$0.0455	0.309 [0.273, 0.345]	\$0.0475	–
qwen3-235b-thinking	telecom	114	0.649	\$0.0866	0.240 [0.205, 0.276]	\$0.1063	–
qwen3-235b-thinking	AVERAGE	92.7	0.675	\$0.0615	0.272 [0.228, 0.314]	\$0.0695	–
qwen3-32b	airline	50	0.100	\$0.0252	0.056 [0.012, 0.106]	–	–
qwen3-32b	retail	114	0.561	\$0.0312	0.282 [0.235, 0.329]	\$0.0373	–
qwen3-32b	telecom	114	0.263	\$0.0416	0.126 [0.089, 0.165]	–	–
qwen3-32b	AVERAGE	92.7	0.308	\$0.0327	0.155 [0.112, 0.200]	\$0.0373	–
qwen3.5-122b-a10b	airline	50	0.820	\$0.0168	0.471 [0.406, 0.529]	\$0.0149	\$0.0355
qwen3.5-122b-a10b	retail	114	0.807	\$0.0142	0.487 [0.443, 0.532]	\$0.0156	\$0.0311
qwen3.5-122b-a10b	telecom	114	0.991	\$0.0212	0.581 [0.563, 0.597]	\$0.0195	\$0.0294
qwen3.5-122b-a10b	AVERAGE	92.7	0.873	\$0.0174	0.513 [0.470, 0.553]	\$0.0167	\$0.0320
qwen3.5-27b	airline	50	0.860	\$0.0113	0.545 [0.476, 0.607]	\$0.0100	\$0.0170
qwen3.5-27b	retail	114	0.903	\$0.0109	0.589 [0.551, 0.624]	\$0.0105	\$0.0181
qwen3.5-27b	telecom	114	0.991	\$0.0160	0.629 [0.610, 0.645]	\$0.0154	\$0.0213
qwen3.5-27b	AVERAGE	92.7	0.918	\$0.0127	0.588 [0.546, 0.625]	\$0.0120	\$0.0188
qwen3.5-35b-a3b	airline	50	0.580	\$0.0069	0.410 [0.313, 0.505]	\$0.0085	–
qwen3.5-35b-a3b	retail	114	0.798	\$0.0070	0.583 [0.526, 0.640]	\$0.0071	–
qwen3.5-35b-a3b	telecom	114	1.000	\$0.0104	0.705 [0.691, 0.718]	\$0.0102	\$0.0133
qwen3.5-35b-a3b	AVERAGE	92.7	0.793	\$0.0081	0.566 [0.510, 0.621]	\$0.0086	\$0.0133
qwen3.5-397b-a17b	airline	50	0.820	\$0.0244	0.424 [0.361, 0.476]	\$0.0208	\$0.0433
qwen3.5-397b-a17b	retail	114	0.825	\$0.0208	0.452 [0.413, 0.490]	\$0.0217	\$0.0385
qwen3.5-397b-a17b	telecom	114	1.000	\$0.0291	0.526 [0.514, 0.538]	\$0.0294	\$0.0371
qwen3.5-397b-a17b	AVERAGE	92.7	0.881	\$0.0248	0.467 [0.429, 0.501]	\$0.0239	\$0.0396
qwen3.6-flash	airline	50	0.800	\$0.0238	0.411 [0.349, 0.468]	\$0.0215	\$0.0448
qwen3.6-flash	retail	114	0.825	\$0.0202	0.454 [0.415, 0.492]	\$0.0204	\$0.0311
qwen3.6-flash	telecom	114	1.000	\$0.0298	0.522 [0.510, 0.534]	\$0.0294	\$0.0371
qwen3.6-flash	AVERAGE	92.7	0.875	\$0.0246	0.462 [0.425, 0.498]	\$0.0238	\$0.0377
qwen3.7-max	airline	50	0.820	\$0.1993	0.160 [0.133, 0.187]	\$0.2009	\$0.3786
qwen3.7-max	retail	114	0.746	\$0.1753	0.143 [0.125, 0.162]	\$0.1843	–
qwen3.7-max	telecom	114	0.974	\$0.2628	0.143 [0.131, 0.155]	\$0.2633	\$0.3327
qwen3.7-max	AVERAGE	92.7	0.846	\$0.2124	0.149 [0.130, 0.168]	\$0.2162	\$0.3557
qwen3.7-plus	airline	150	0.847	\$0.0349	0.385 [0.356, 0.413]	\$0.0332	\$0.0625
qwen3.7-plus	retail	342	0.792	\$0.0309	0.381 [0.360, 0.402]	\$0.0311	–
qwen3.7-plus	telecom	340	0.985	\$0.0440	0.447 [0.438, 0.455]	\$0.0417	\$0.0559
qwen3.7-plus	AVERAGE	277.3	0.875	\$0.0366	0.404 [0.384, 0.423]	\$0.0353	\$0.0592